The Latest In SENSORS AND APPLICATIONS

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The new wave of sensors, including those used in IoTs and wearables, are soon going to revolutionise electronics industry

Be it a silent heart attack detector that detects the protein level of a patient or a posture-correcting chair that alerts the occupant sitting in a wrong posture—both recently invented by Indian teenagers—sensors have a vital role to play in electronic devices. The fact is that the application of sensors is ever-expanding along with the progress in science and technology.

As per industry reports, sensors are becoming the biggest and fastest growing markets, comparable with computers and communication devices markets. You find sensors in smartphones, automobiles, security systems and even everyday objects like coffee makers! Apart from consumer electronics, these are also an integral part of the Internet of Things (IoT), medical, nuclear, defence, aviation, robotics and artificial intelligence, agriculture, environment monitoring and deep-sea applications.

The shift to smarter sensors

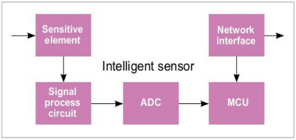
Basically, a sensor is an input device that receives and responds to a signal or stimulus. Nowadays some sensors come integrated with many sensing elements and read-out circuitry in a single silicon chip, providing high accuracy and multiple functions. Manufacturers use both advanced technologies and methods for signal processing and conversion. Modern sensors have more features including user-friendliness, accessibility and flexibility. So there is a paradigm shift in the sensor industry with integration of new technologies to make sensors smarter and intelligent.

Ordinary sensors are still used in many applications. But innovation and advancement in microelectronics is taking sensor technology to a completely new level. The functionality of ordinary sensors has expanded in many ways and these now provide a number of additional properties. Sensors are becoming more and more intelligent, providing higher accuracy, flexibility and easy integration into distributed systems.

Intelligent sensors use standard bus or wireless network interfaces to communicate with one another or with microcontrollers (MCUs). The network interface makes data transmission easier while also expanding the system. Manufacturers can diagnose sensor faults and guide users to troubleshoot them remotely through the computer network.

An intelligent sensor may consist of a chain of analogue and digital blocks, each of which provides a specific function. Data processing and analogue-to-digital conversion (ADC) functionalities help improve sensor reliability and measurement accuracy. The typical structure of an intelligent sensor is shown in Fig. 1.

Fig. 1: Intelligent sensor structure (Courtesy: www.mdpi.com)



Common types of latest sensors

There are a wide variety of sensors depending on the technology (analogue/digital) and applications. This article covers some of the latest sensors including IoT sensors, pollution sensors, RFID sensors, image sensors,

Market size

According to a BBC Research report, the global sensor market is chasing double-digit growth. It expects the global market for sensors to reach $240.3 billion in 2021, up from about $123.5 billion in 2016. Fingerprint sensors will lead the market by growing at 15.9 per cent annually.

Another source puts chemical sensors, process variable sensors, proximity and positioning sensors among the fastest growing markets. Automotive industry will once again be the leading consumer market for sensors.

A TechSci Research report says, "India's sensors market is one of the fastest growing markets in Asia-Pacific. Rising security concerns and growing trend towards miniaturisation are shifting the focus of consumers to smart devices, which make extensive use of various sensors, in particular, touch and image sensors." The report projects India's sensors market to grow at over 20 per cent annually during 2015-20.